

Question:
What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>

Key idea: For low-speed electrons, the kinetic energy relation is $K \approx \frac{1}{2} m v^2$, so de Broglie wavelength $\lambda = h / p = h / (m v) \approx h / \sqrt{2 m K}$.

- Electron rest mass $m \approx 9.11 \times 10^{-28}$ g

- 1 eV $\approx 4.135 \times 10^{-12}$ erg

- Kinetic energy $K = 1$ eV $= 4.135 \times 10^{-12}$ erg

- 1 eV corresponds to frequency $\nu = E/h = (4.135 \times 10^{-12} \text{ erg}) / (6.625 \times 10^{-27} \text{ erg}\cdot\text{s}) \approx 6.24 \times 10^{14} \text{ s}^{-1}$

- Wavelength $\lambda = c / \nu \approx 3.00 \times 10^{10} \text{ cm/s} / 6.24 \times 10^{14} \text{ s}^{-1} \approx 4.81 \times 10^{-5} \text{ cm}$

- Alternatively, de Broglie: $h/\lambda = p = \sqrt{2 m K}$, giving $\lambda = h / \sqrt{2 m K}$; using $K = 4.135 \times 10^{-12}$ erg and $m = 9.11 \times 10^{-28}$ g yields $\lambda \approx 4.14 \times 10^{-10} \text{ m} = 4.14 \times 10^{-5} \text{ cm}$.

Thus, the de Broglie wavelength is about 4.14×10^{-5} cm, which corresponds to choice D.

</think>

<answer>

D. 3.14×10^{-5} cm

</answer>